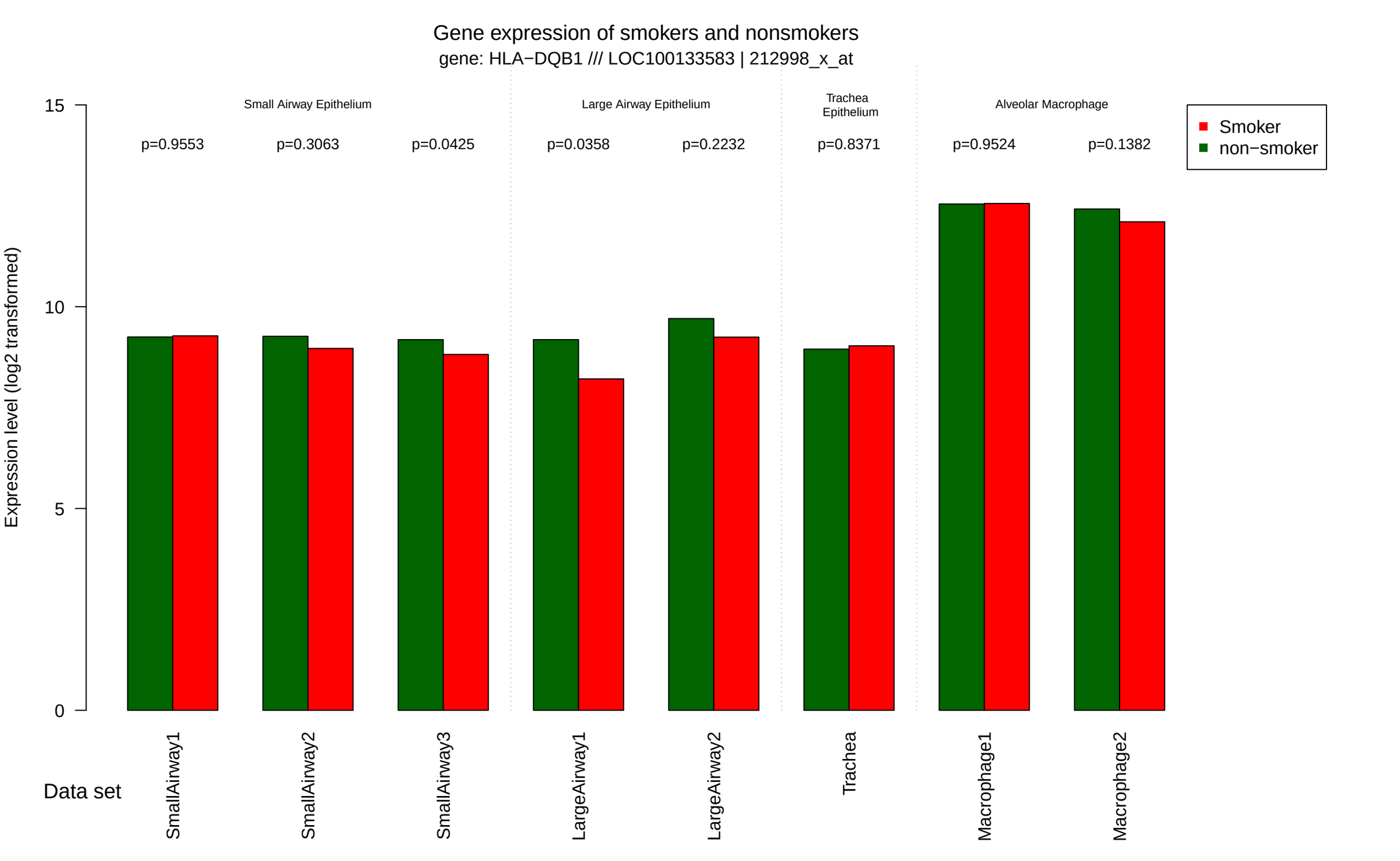
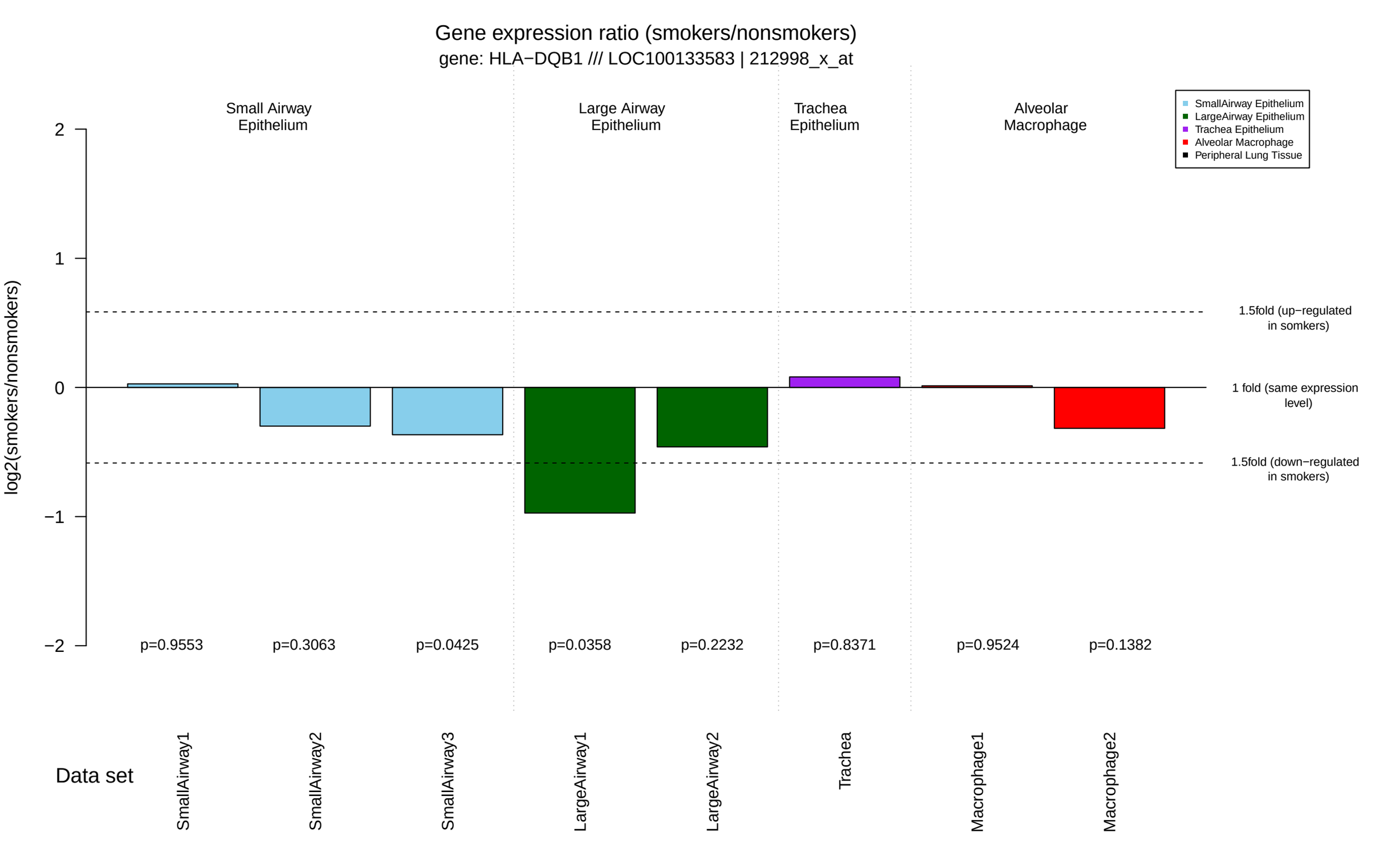
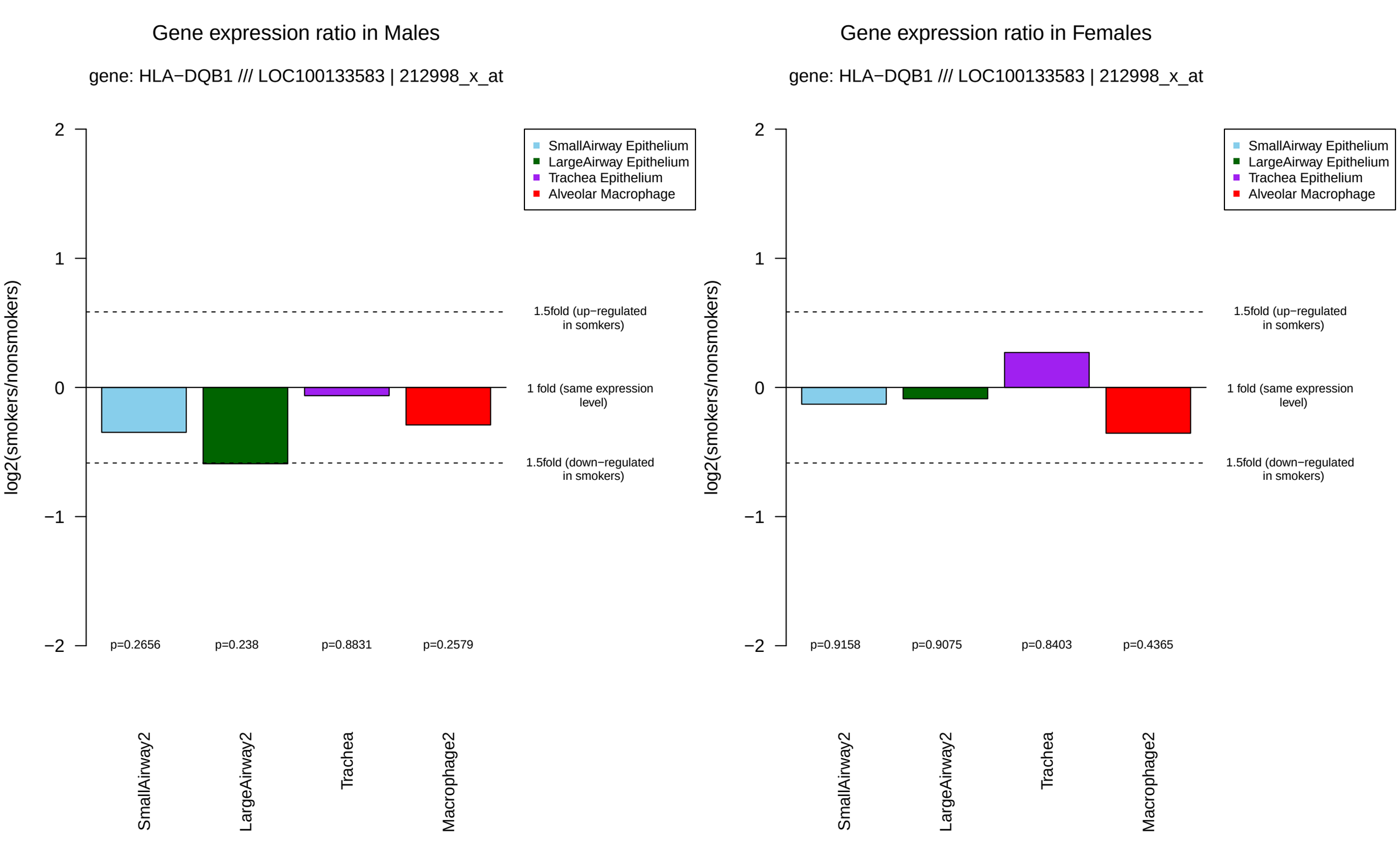
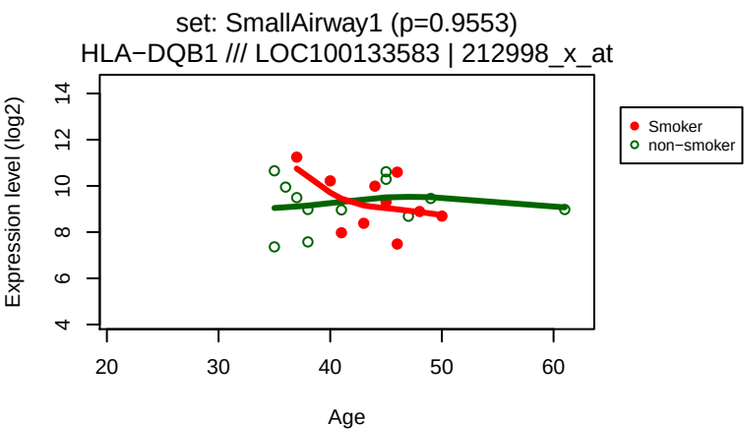
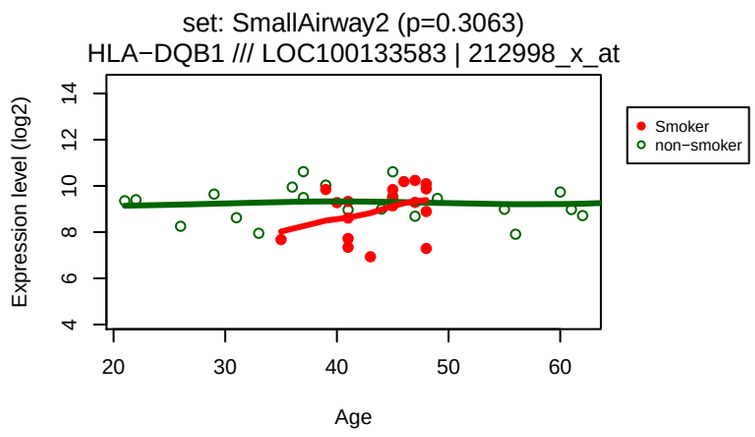
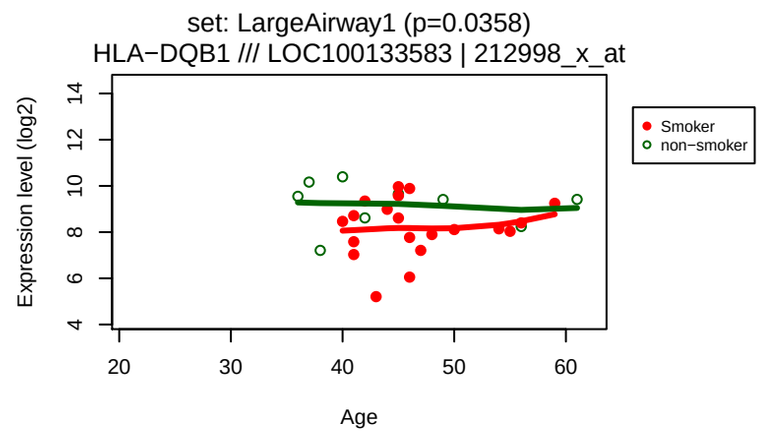

Gene expression of smokers and nonsmokers
gene: HLA−DQB1 /// LOC100133583 | 212998_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9553
p=0.3063
p=0.0425
p=0.0358
p=0.2232
p=0.8371
p=0.9524
p=0.1382
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

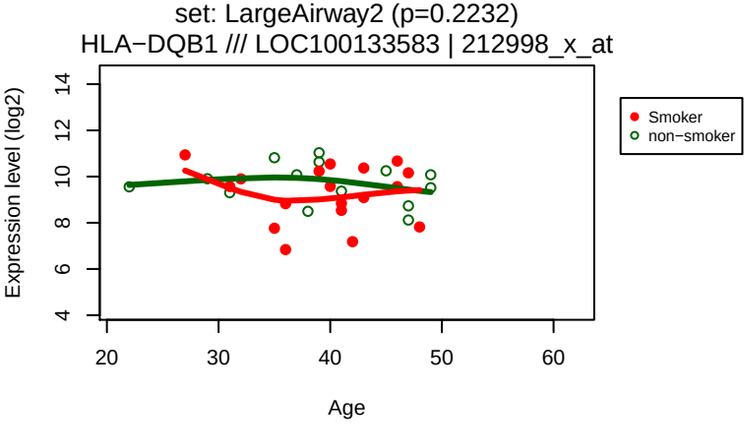

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DQB1 /// LOC100133583 | 212998_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9553
p=0.3063
p=0.0425
p=0.0358
p=0.2232
p=0.8371
p=0.9524
p=0.1382
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

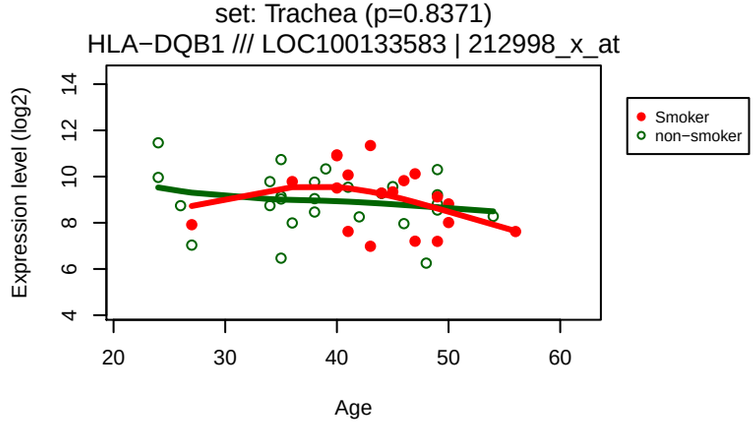

Gene expression ratio in Males
gene: HLA−DQB1 /// LOC100133583 | 212998_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2656
p=0.238
p=0.8831
p=0.2579
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DQB1 /// LOC100133583 | 212998_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9158
p=0.9075
p=0.8403
p=0.4365
SmallAirway2
LargeAirway2
Trachea
Macrophage2

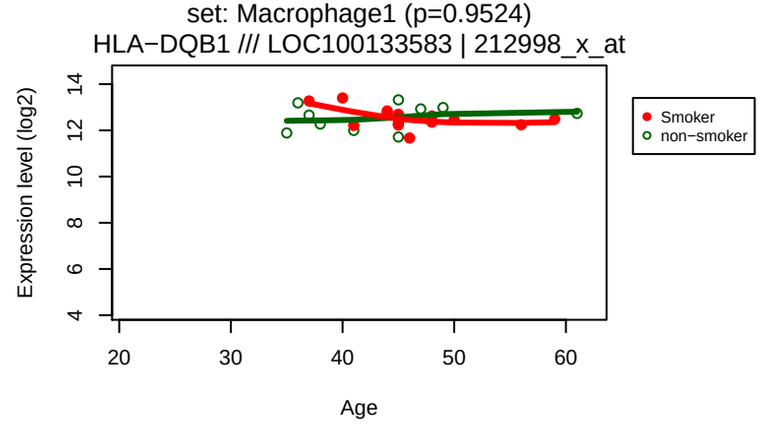

set: SmallAirway1 (p=0.9553)
HLA−DQB1 /// LOC100133583 | 212998_x_at
Expression level (log2)
Age
Smoker
non−smoker

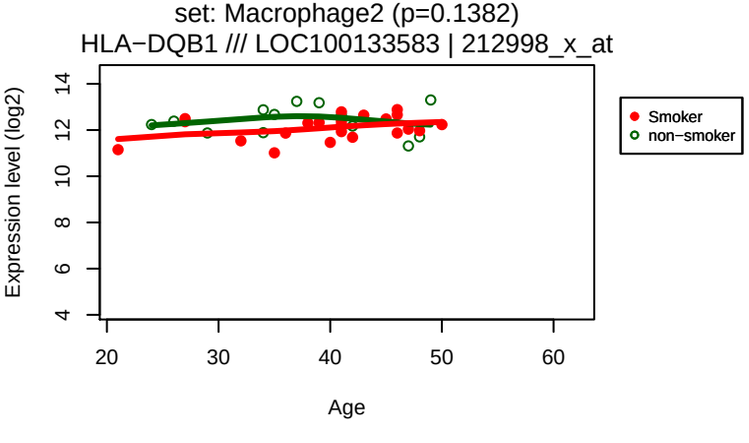

set: SmallAirway2 (p=0.3063)
HLA−DQB1 /// LOC100133583 | 212998_x_at
Expression level (log2)
Age
Smoker
non−smoker

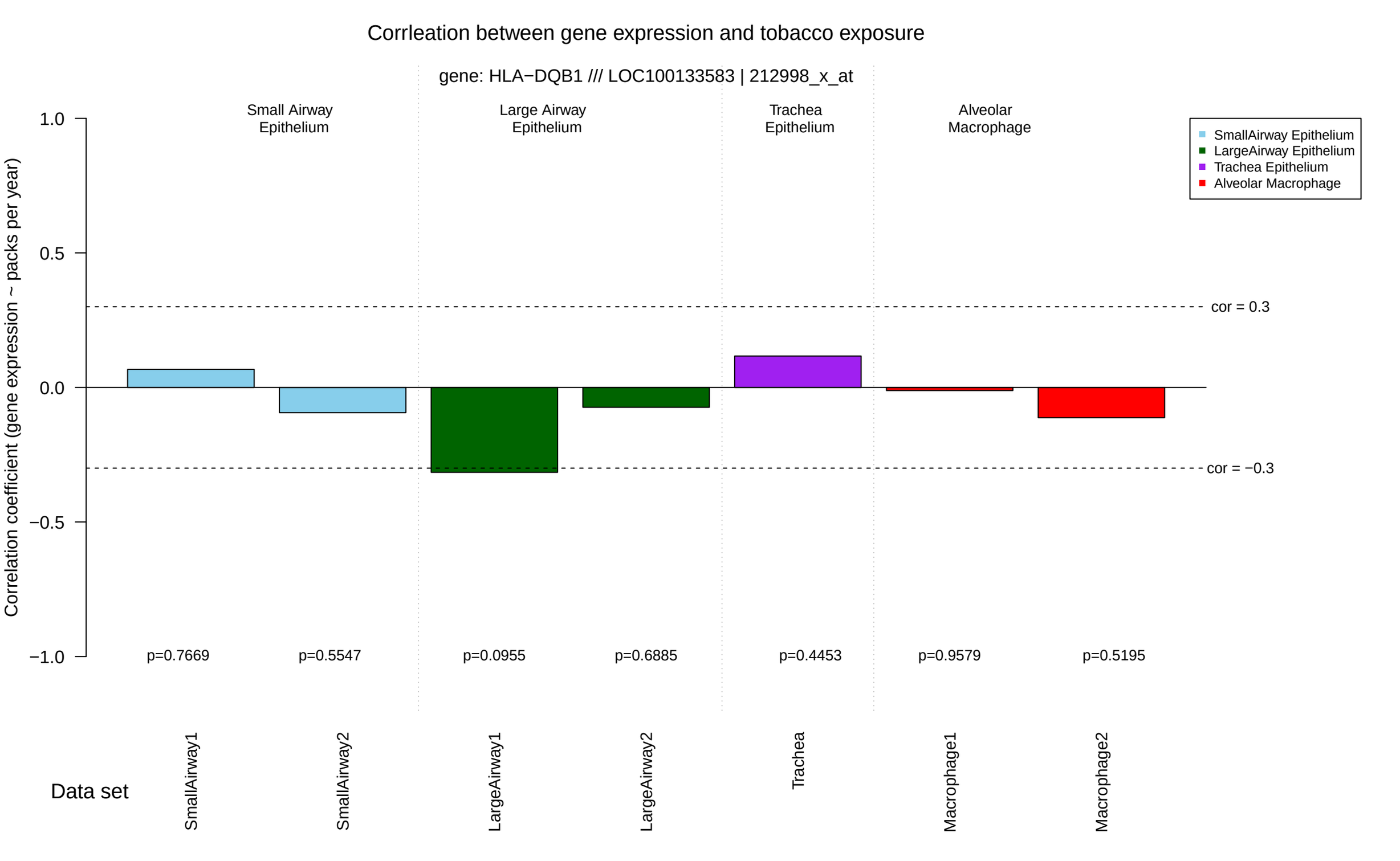

set: LargeAirway1 (p=0.0358)
HLA−DQB1 /// LOC100133583 | 212998_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2232)
HLA−DQB1 /// LOC100133583 | 212998_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8371)
HLA−DQB1 /// LOC100133583 | 212998_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9524)
HLA−DQB1 /// LOC100133583 | 212998_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1382)
HLA−DQB1 /// LOC100133583 | 212998_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DQB1 /// LOC100133583 | 212998_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7669
p=0.5547
p=0.0955
p=0.6885
p=0.4453
p=0.9579
p=0.5195
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2